



EventScape

Mobile and Web based Event Layout Planner with 3D Visualization and Preview

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

INTRODUCTION

Event planning has evolved significantly in recent years, driven by rapid technological advancements and growing consumer expectations worldwide. On a global scale, the event planning industry has increasingly shifted toward digital solutions, with the adoption of virtual and hybrid event formats, cloud-based coordination tools, and immersive design technologies becoming standard practice among professional planners. The global event management software market was estimated at USD 15.5 billion in 2024 and is projected to reach USD 34.7 billion by 2029, growing at a CAGR (Compound Annual Growth Rate) of 17.4%, reflecting the rising demand for more efficient, technology-driven approaches to organizing events of all scales (MarketsandMarkets, 2025). Central to this shift is the use of 3D visualization technology, which allows planners to create realistic and immersive representations of venues and layouts before any physical setup begins. Advanced 3D modeling enables planners to produce virtual representations of venues, layouts, and stage setups prior to construction, improving coordination and significantly reducing errors (Fannlow, 2025). Traditional 2D event layouts, while still functional, often fail to fully capture the spatial dynamics, scale, and overall ambiance of a venue, which has further accelerated the adoption of 3D event design solutions that offer a more realistic and comprehensive visualization experience (Bata, 2024).

Despite these advancements, event planning continues to be a complex and time-consuming process for many organizers around the world. Manual planning methods, including paper-based layouts and in-person coordination, consume considerable time and are prone to communication breakdowns, scheduling conflicts, and budget overruns. These inefficiencies frequently result in client dissatisfaction, financial miscalculations, and operational failures that could have otherwise been avoided (Kumar, 2025). At the local level, the situation is even more pronounced. Event planning in many local settings remains largely traditional, characterized by minimal client participation, limited collaborative tools, and a heavy reliance on physical coordination. The absence of AI-assisted design features, interactive 3D previews, and integrated scheduling tools continues to reduce overall efficiency and engagement for both planners and their clients.

Several existing systems have already demonstrated the potential of technology in addressing some of these concerns. Prismm, for instance, offers tools for 3D event design, seating arrangements, and multi-user collaboration, while 3D Event Designer provides interactive 2D and 3D layout planning and visualization capabilities. However, these platforms are primarily designed for professional planners rather than end clients, and they notably lack integrated AI-assisted design features, automatic cost estimation, and dedicated scheduling tools capable of preventing venue conflicts. These gaps highlight the continued need for a more comprehensive and client-centered solution.



To address these limitations, the proposed Mobile and Web-based Event Layout Planner with 3D Visualization and Preview aims to enhance the event planning experience through an integrated and inclusive approach. The system enables users to create and modify event layouts using both 2D and 3D design tools, ready-made venue templates, and AI-assisted design features. It further supports 3D previews, interactive layout adjustments, active customer participation in design decisions, and automatic cost estimation. A venue scheduling feature is also incorporated to ensure efficient space utilization and prevent booking conflicts. By combining these capabilities into a single platform, the system seeks to simplify the event planning process, strengthen collaboration between organizers and clients, and promote more informed decision-making throughout every stage of event preparation.

GENERAL OBJECTIVES OF THE STUDY

The general objective of this study is to develop a mobile and web-based event layout planner using 2D and 3D visualization to improve the efficiency, accuracy, and reliability of event planning.

Specifically, this study aims to address the following statements:

1. To develop a module that facilitates layout planning, venue management, and scheduling.
 - 1.1. To develop a user-friendly interface for planning and customizing event layouts in 2D and 3D.
 - 1.2. To design a module for placement and arrangement of event elements, including chairs, tables, stages, and other event elements.
 - 1.3. To provide tools for managing venues and sharing or exporting layouts.
 - 1.4. To implement AI-assisted design and cost computation features to enhance the visual of event layout and support budget management.
 - 1.5. To develop a scheduling module to manage event bookings, track venue availability, and prevent conflicts.
2. To develop a module that generates insights and reports to support informed decision-making in event management.
 - 2.1 To generate reports on event layouts and arrangements.
 - 2.2 To provide analytics on venues and applied templates.
 - 2.3 To produce booking and scheduling reports to track events and prevent conflicts.

3. To evaluate and assess the system using a standard evaluation tool as perceived by end-users and IT Experts.

SIGNIFICANCE OF THE STUDY

This study is significant because it provides tangible benefits to multiple stakeholders involved in event planning and execution by offering a web-based platform for administrators and a mobile application for users to design and interact with event layouts.

Clients and Stakeholders. Clients can interact with interactive 3D previews of event layouts, including seating arrangements, stage placement, décor, and overall flow. This facilitates informed decision-making, accelerates approvals, and aligns client expectations with the implemented setup, improving satisfaction and engagement.

Educational Institutions and Researchers. Students and researchers in information technology, event management, and multimedia design may reference this study to understand user-centered design, 3D visualization techniques, and digital tools for event management. It contributes to academic knowledge on interactive planning systems and practical design methodologies.

Event Organizers. The system enables organizers to design and visualize event layouts in a virtual 3D environment, allowing adjustments that minimize planning errors and optimize space utilization, while organizers can efficiently manage schedules and prevent conflicts.

Future System Developers. The study provides insights into system architecture, user interface design, and 3D rendering integration. It serves as a foundation for future enhancements, such as augmented reality (AR), virtual reality (VR), expanded AI-assisted design and layout guidance, and cost estimation modules.

SCOPE AND DELIMITATIONS/LIMITATIONS OF THE STUDY

This study focuses on the design and development of a mobile and web-based Event Layout Planner with 3D Visualization and Preview, which assists event organizers, affiliated venues, and clients in planning and designing event layouts. The system enables users to create and customize event layouts in 2D / 3D and view them through 3D previews. Users may use ready-made venue templates or create custom layouts by inputting venue dimensions, door placements, and other structural elements. The system includes AI-assisted design features that allow users to generate visually appealing layouts by applying user-defined prompts for color schemes, décor elements, and overall visual themes while keeping the design suitable for the intended event function. It also includes a wall-extraction service using OpenCV and NumPy for deterministic image preprocessing, wall mask cleanup, line extraction, opening detection, and wall candidate generation to identify walls, doors, and windows from uploaded floorplan images. The system also use a local Ultralytics YOLO object-detection model to provide supporting semantic hints for floorplan elements such as walls, doors, windows, columns, stairs, and other structural features. Final walls are then generated through deterministic validation, graph cleanup, review layers, and user confirmation to ensure that the output remains editable, reviewable, and suitable for 2D and 3D venue visualization. On-device camera scan quality is validated locally using MobileNetV3 before server-side processing. Metadata and OCR text extracted from blueprint scans are normalized using Mistral 7B, a lightweight open-weight

large language model, to produce clean and structured JSON output. Scheduling conflict detection and layout validation are handled by classical machine learning models, specifically XGBoost and RandomForest, which process structured inputs such as guest count, venue area, and asset configurations to flag conflicts and predict setup requirements. The system also supports basic cost computation, venue scheduling, and layout export in PDF format.

The mobile application is intended for Event Organizers or Staff, Affiliated Venues or Venue Administrators, and Event Company Customers or Clients. Meanwhile, the web application is intended for Event Company Admins, who have full oversight and management access. Both platforms require a stable internet connection, as the system does not support offline functionality. The system is designed specifically for event layout visualization and design and does not cover comprehensive event planning, program sequencing, or task coordination. The system is limited to pre-defined venue templates and custom layouts based on user-provided venue details. It does not support importing external file types, and layout exports are limited to Images and PDF format only. Mobile access is restricted to Android and iOS devices, while web access is available exclusively to Event Company Admins. Scheduling capabilities are limited to date selection and venue assignment to prevent double-booking and do not include the sequencing or management of event activities.

Each AI component within the system is limited to its designated function and does not replace professional event planning or architectural expertise. The wall-extraction service is limited to detecting structural elements such as walls, doors, and windows from uploaded floorplan images and may not accurately identify all architectural features or complex structural configurations. Mistral 7B is used only for OCR text cleanup and metadata normalization and is not intended for generative or creative design purposes. MobileNetV3 is

limited to validating camera scan quality on-device and does not process or interpret layout content. The XGBoost and RandomForest models handle conflict detection and layout validation based strictly on structured inputs such as guest count, venue area, and asset configurations. These models do not account for real-world variables outside the system's parameters, such as physical obstructions, sudden event changes, or regulatory compliance. Automated validation is limited to seating capacity, sightlines, and flow efficiency and does not assess structural safety standards or building code compliance. Third-party integrations such as CAD software, lighting systems, acoustics modeling, and structural simulations are not supported. The system also does not support collaborative editing across multiple devices. Camera-assisted scaling, including optional monocular depth estimation, provides approximate venue measurements only and should not be used as a substitute for professional or architectural assessment. System performance may vary depending on device specifications, and regular updates are required to maintain the accuracy of templates, validation rules, and export functions. Overall, the system serves as a practical tool for event layout visualization, design assistance, and basic venue scheduling. It is not intended to replace comprehensive event planning, professional design consultation, architectural assessment, or operational event management.

DEFINITION OF TERMS

2D Visualization – A two-dimensional scaled drawing of a building level seen from above that communicates the spatial organization of a space, including walls, doors, windows, and other physical features (Pizarro et al., 2022). In this study, 2D visualization refers to the top-down floor plan view where users can drag and drop event elements like tables and chairs.

3D Visualization - Creation of three-dimensional representations of objects or environments using 3D software, allowing realistic and interactive viewing (Arena, 2025). As used in this study, 3D visualization refers to the system feature that lets users view and interact with event layouts in 3D.

AI-Assisted Design - Workflows use artificial intelligence to automate repetitive design tasks, generate initial drafts of layouts or content, and support faster iteration while augmenting, rather than replacing human designers (Brummer, 2025). As used in this study, AI-assisted design refers to a system feature that applies to user-defined prompts to automatically design the visual appearance of an event layout, including colors, décor, and overall aesthetic.

Real-time Rendering – A computer graphics process that generates and processes images quickly enough (at least 30 frames per second) to enable interactivity, allowing users to see immediate visual feedback as they modify a 3D scene (Kim & Chai, 2020). In this study, it refers to the system's ability to instantly update the 3D preview as the user makes changes in the 2D layout.

Venue Template – A pre-designed digital layout stored in an event management system that can be reused or customized to manage different event spaces and reduce repetitive setup time (Shah et al., 2023). In this study, these are ready-made room dimensions and structures (including door and window placements) that users can choose to save time during the planning process.

Venue Management – The technology-enabled process of administering event spaces, which includes mobile and web applications that allow organizers to handle venue information, monitor space usage, and coordinate event logistics (García Revilla et al., 2023). As used in this study, it refers to the module that allows administrators to input venue dimensions, upload structural templates (door and window placements), and monitor the utilization of specific event spaces.

Scheduling Module – A software system that manages room or space availability by integrating real-time information into a booking application to control utilization and prevent scheduling conflicts (Tran et al., 2016). In this study, it refers to the tool used to manage venue availability and prevent double-booking of event spaces.

CHAPTER II

REVIEW OF RELATED LITERATURE/ STUDIES

The following are the literature, studies and existing applications that provide the base for the idea of the Mobile and Web-Based Event Layout Planner with 3D Visualization and Preview, as well as the foundation for the development of it.

RELATED LITERATURE

This section discusses the theoretical foundations relevant to the development of the Mobile and Web-Based Event Layout Planner with 3D Visualization and Preview.

Centralized Event Creation, Booking, and Scheduling Management

Traditional event management frameworks are heavily reliant on legacy manual tracking, leading to systemic operational inefficiencies such as booking overlaps, coordination conflicts, and administrative processing delays. To address these operational challenges, Capuno et al. (2021) introduced centralized online booking matrices, algorithmic conflict-free scheduling engines, and automated communication channels to reduce communication latency and secure initial scheduling data. Reinforcing this transition, Agpalo, Dela Cruz, and Soriano (2025) demonstrated that real-time tracking modules optimize day-to-day workflow coordination and maintain accurate records across administrative environments. Furthermore, Shah, Vasudavan, and Razali (2023) highlighted the technical value of replacing fragmented event logistics with centralized Event Management Systems (EMS) to eliminate information silos and ensure that real-time data updates are instantly accessible to all connected users.

This literature is aligned with the backend scheduling features and operational workflows of EventScape: Mobile and Web-based Event Layout Planner with 3D Visualization and Preview. By demonstrating that scheduling automation and database centralization stabilize the logistics pipeline, these studies academically support EventScape's unified database framework. This justifies the development of an advanced venue scheduling engine that prevents booking conflicts as a necessary administrative foundation before spatial layout modeling can safely take place.

Performance Integrity inside Centralized Web and Mobile Environments

Early digital event planning utilities functioned primarily as static administrative portals, offering limited visual support for long-distance spatial organization. JosephNg, Tan, and Lim (2022) explained that text-heavy administrative systems often cause communication delays, emphasizing the need to integrate advanced visual features into centralized platforms to improve logistical coordination. To support this functionality within browsers, Deng et al. (2022) demonstrated that WebGL and interactive web-based 3D architectures enable efficient data binding and real-time rendering directly through standard web interfaces with minimal latency. Furthermore, Radu (2023) explained that cloud-centric data structures optimize deployment speed and reduce human error by linking operational parameters directly with spatial layouts, thereby consolidating administrative and design data into a unified digital ecosystem.

This literature highlights the structural feasibility and technical requirements needed to host interactive media within online systems. It directly supports one of EventScape's main technical objectives: establishing synchronization and performance stability between a mobile application and a web-based management dashboard. These studies demonstrate that modern hardware-accelerated engines can effectively combine business records, centralized data, and 3D preview interfaces within standard consumer operating environments.

Interactive 3D Visualization and Spatial Layout Optimization

Traditional venue planning methodologies often struggle to convey spatial dimensions accurately, resulting in layout errors and manual adjustments during physical setup. Phoaroon, Boonsri, and Taylor (2025) established that integrating immersive 3D layout tools and interactive spatial configurations enhances layout comprehension, improves multi-stakeholder communication, and minimizes layout errors by addressing the structural limitations of flat drawings. Supporting this need for multi-dimensional configuration, Safikhani et al. (2022) demonstrated that flat 2D blueprints require high levels of mental abstraction, while immersive 3D visualization reduces cognitive load and helps prevent physical asset collisions that may be overlooked in flat environments. Additionally, Rubio-Sandoval, Fernandez, and Jimenez (2021) emphasized that optimized, lightweight cross-platform rendering engines are important for maintaining smooth performance and reducing application latency, allowing users to identify structural placement conflicts in real time.

This literature directly reinforces the core visual module of EventScape: Mobile and Web-based Event Layout Planner with 3D Visualization and Preview. The literature shows that traditional 2D blueprints can create cognitive demands and increase the risk of setup flaws, while 3D visualization engines improve spatial understanding and layout validation. This supports EventScape's 2D/3D dual-interactive layout planner by justifying the need for responsive asset manipulation, real-time previewing, and visual conflict identification in event layout design.

Immersive Environments, Spatial Perception, and Layout Accuracy

Moving beyond two-dimensional layout boundaries significantly enhances how users perceive physical constraints and recall environmental details. Behmanesh and Brown (2025) validated the practical utility of this approach by demonstrating that interactive 3D visualizations allow layout designers to assess flow patterns, optimize spatial geometry, and identify physical bottlenecks in complex public spaces. Supporting this cognitive shift, Proulx and Çöltekin (2025) explained that realistic visual depth enhances user comprehension by providing stronger spatial perception and clearance assessment compared to flat graphics. Similarly, Wiltenburg, van der Meer, and de Vries (2024) established that interactive modeling alters cognitive scaling, allowing non-technical users to use egocentric perspective-taking to identify physical bottlenecks and improve overall spatial memory.

This literature directly anchors EventScape's automated verification tools and 3D preview engine. The literature provides academic support that moving beyond traditional 2D limitations can help the system evaluate seating capacity, sightlines, and flow efficiency more effectively. It also supports EventScape's role as a predictive planning tool that reduces the risk of physical setup mismatches, material waste, and on-site layout adjustments.

Fostering Stakeholder Engagement and Communicating Design Intent

A recurring obstacle in remote layout design is accurately conveying abstract spatial concepts to non-technical clients, which may cause misunderstanding and post-implementation concerns before physical setup begins. Canizaro (2024) explored the media and tools used in architectural visualization, emphasizing that sensory-rich and active user participation within 3D virtual environments helps bridge communication gaps and allows non-technical users to better comprehend complex spatial arrangements. Expanding on this, Lee, Kim, and Park (2023) demonstrated that translating abstract design indicators into immersive visual setups strengthens stakeholder trust and engagement by reducing discrepancies in design intent. Furthermore, Chen and Harris (2022) documented that real-time 3D walkthroughs help reduce client friction by setting shared design expectations before physical setup, thereby streamlining the remote approval process across standard tablets or smartphones.

This literature aligns directly with the client-centered design goals and interactive preview features of EventScape: Mobile and Web-based Event Layout Planner with 3D Visualization and Preview. By showing that visual walkthroughs can reduce post-implementation client friction, these studies support EventScape's collaborative model. The literature also justifies the integration of cost computations, visual templates, and 3D preview walkthroughs into a single platform, as these features help address communication gaps among event companies, coordinators, and non-technical clients.

RELATED STUDIES

This section reviews related empirical studies and existing systems that support the development of the Mobile and Web-Based Event Layout Planner with 3D Visualization and Preview.

AI-Driven 2D-to-3D Vector Floor Plan Generation

Automated floor plan generation has shifted toward creating complex spatial layouts by arranging structural elements while balancing topological, geometric, and aesthetic considerations. Zhuang et al. (2026) introduced controllable end-to-end vector floor plan generation models that bypass traditional multi-step drawing processes, allowing AI to learn floor plan creation directly from vector data without predefined human pathways. Supporting this trajectory, Albano, Liu, Wan, and Li (2026) demonstrated that this transition from manual analog modeling to machine learning significantly enhances the speed and structural logic of

spatial abstractions. Chan (2026) further established that physical-first, multi-modal workflows seamlessly transcode flat 2D blueprint geometries into 3D forms, while Stojaković and Đurić (2025) confirmed that extracting topological features from segmentation maps is essential for maintaining coherence in automated layout generation.

This literature provides an authoritative foundation for EventScape's 3D Rendering Core, which utilizes the Three.js engine and React Three Fiber. By confirming that end-to-end vector extraction successfully drives spatial design, these studies academically justify EventScape's deterministic geometry engine. It proves that converting 2D pixel layouts into topology-safe vector lines allows the system to instantly and dynamically extrude flat venue coordinates into a highly accurate, interactive 3D layout environment.

Convolutional Neural Networks and YOLO in Structural Blueprint Extraction

Traditional architectural symbol spotting often struggles with graphical notation variability, necessitating robust deep learning interventions. Kim et al. (2024) demonstrated that upgrading floor plan object detection to single-stage detectors, specifically the YOLO framework, drastically reduces training and inference time while achieving superior localization accuracy for spatial objects like walls and fixtures. Reinforcing this technical approach, Rezvanifar (2021) established that deep learning methods overcome the high density of information in 2D drawings. Furthermore, Anand and Bhattacharya (2024) confirmed that lightweight, on-device networks successfully pre-screen image quality on mobile hardware, while Dos Santos, Silva, and Mendes (2024) verified that convolutional networks facilitate the automated extraction of boundaries into structured database schemas.

This literature directly supports the computer vision infrastructure of EventScape. The academic validation of YOLO-based object detection justifies EventScape's use of Ultralytics YOLO11m as a server-side vision engine for identifying structural elements such as walls, doors, and windows from uploaded floorplans. Furthermore, studies on lightweight on-device models support the system's use of MobileNetV3 for local camera scan validation, helping ensure that physical venue constraints are properly translated into the system's interactive 2D/3D dual layout canvas.

Text-to-Design Generative AI in Interior and Spatial Aesthetics

Generative AI has significantly reshaped design workflows by enhancing creative freedom, visualization efficiency, and user participation in the design process. Gökdemir and Kayan (2025) examined the reconfiguration of AI-driven design tools within interior design education, emphasizing their role as linguistic interfaces that mediate cognitive and communicative processes in design. Their study, published in the *International Journal of Architectural Computing*, analyzed an AI-based workshop with interior design students and found that natural language prompts allow users to express design intent, which AI systems translate into structured design outputs. This demonstrates a shift from traditional manual design methods toward a linguistically embedded computational approach, where users without deep technical expertise can meaningfully participate in design creation.

This is supported by Omer, Desai, and Patel (2024), who demonstrated that generative AI empowers users to conceptualize designs more vividly, while Zhang and Garcetti (2024) provided evidence that user-defined text prompts can effectively drive algorithmic style sheets.

Ahmed and Alymani (2024) also showed that AI-driven rendering assistants facilitate comparative visual studies, significantly improving the translation of abstract creative concepts into cohesive layout themes. At the output level, Chen, Shao, Zheng, Zhang, and Yin (2024) further reinforced this by demonstrating that text descriptions specifying decoration styles and spatial functions can be used in AI diffusion models to efficiently generate aesthetically pleasing interior design results. Their findings also confirmed that modifying the text description allows rapid regeneration of new designs, proving that natural language is an effective medium for controlling spatial and aesthetic outputs.

This literature supports EventScape's AI-Assisted Design and visual template feature. The studies confirm that text-driven generative tools are effective in customizing thematic and visual elements based on user intent. This justifies EventScape's module that allows clients to use user-defined prompts to automatically style the visual appearance, color schemes, and decor motifs of an event layout, transforming basic 2D wireframes into visually appealing presentations.

Predictive Machine Learning for Spatial Validation and Logistics

Beyond aesthetics, successful spatial management requires predictive machine learning models to evaluate fitness functions such as circulation and structural efficiency. Alqahtani, Siddiqui, and Alghamdi (2024) demonstrated that optimizing space utilization requires models that process quantitative variables to predict optimal arrangements. Supporting the efficacy of this approach, Henderson and Kapoor (2025) proved that classical tree-based models provide

superior accuracy when validating constraints. Furthermore, Thompson and Nguyen (2023) verified that these classifiers effectively evaluate aisle clearance and sightlines, while Garcia and Nielsen (2024) confirmed that linking machine learning conflict detectors with centralized operational records drastically reduces financial and logistical errors during the design phase.

This literature supports the separation of EventScape's AI models according to their intended functions. Rather than relying on text-based models for calculations, EventScape utilizes tree ensemble models such as XGBoost and RandomForest for structured numerical processing. The studies show that classical tabular machine learning models are well suited for handling features such as guest count, room area, asset quantity, and pricing data. This supports EventScape's use of these models for budget estimation and layout validation, particularly in identifying potential physical positioning conflicts based on structured system inputs.

Hybrid Human-Machine Collaborative Design Workflows

The most effective spatial applications function via hybrid human-machine collaboration, where AI acts as a decision-support assistant rather than an autonomous replacement. Yilmaz and Cekmis (2025) evaluated the "ArchiGAN" framework, establishing that maintaining human agency within an interactive design loop is crucial for flexible architecture. Supporting this collaboration, Albano, Liu, Wan, and Li (2026) emphasized that allowing users to iteratively adjust AI-generated outputs builds trust and accuracy. Chaillou (2024) demonstrated the necessity of breaking generative tasks into stages for user validation, while Lee, Kim, and Park (2023) highlighted that user-centric design pipelines empower non-technical end users to collaboratively modify AI suggestions.

This literature perfectly reflects EventScape's core ethical and functional design philosophy. EventScape's AI models are bounded decision-support tools that do not substitute for professional expertise. By breaking the system into distinct phases scanning, manual manipulation, and 3D visualization the system ensures the organizer maintains full creative control, augmenting the human process without overriding the user's architectural decisions.

RELATED APPLICATION

This section critically examines existing digital platforms that align with the functional and technical objectives of the proposed Mobile and Web-based Event Layout Planner with 3D Visualization and Preview.

Floorplans by Tripleseat

Floorplans by Tripleseat is a cloud-based application that offers 2D and 3D diagramming for event and space planning. It provides drag-and-drop tools for optimizing spacing and flow, as well as features like shareable links, permissions management, dashboards, and export options to support seamless collaboration among planners. Users can generate photorealistic 3D visuals, multi-angle renderings, and flythroughs, while interactive showrooms allow clients to explore designs and view inventory in 3D across devices. This application demonstrates how digital planning tools can improve efficiency, visualization, and user engagement—principles that can be applied to the development of similar systems.

Prismm

Prismm is a cloud-based platform designed for spatial design and event planning. It enables users to create 2D and 3D floor plans, immersive virtual tours, and dynamic spatial layouts for event spaces. The system incorporates drag-and-drop design tools, libraries of venue floor plans and 3D objects, and collaboration features, allowing planners, clients, and vendors to coordinate within a unified workspace. Photorealistic 3D visualization and digital twin technology support realistic previews of event setups, facilitate client approvals, and reduce the need for onsite visits, thereby improving planning efficiency. The platform also allows sharing of designs via links, embedded views, and interactive tools, enabling stakeholders to explore and modify plans collaboratively. While Prismm excels in visualization and collaborative planning, it does not fully integrate workflow management features such as task scheduling, vendor coordination, and automated budgeting, which are essential for comprehensive event management solutions.

EventDraw

EventDraw is a web-based platform designed to create photorealistic 3D event layouts, providing clients with realistic previews to accelerate approval cycles. The system emphasizes immersive visualization and collaboration, enabling planners to fine-tune seating arrangements, décor, lighting, and staging within a virtual environment. By validating spatial and design decisions in a lifelike 3D setting, the platform helps reduce costly late-stage changes and improve overall planning efficiency. However, while it excels in realistic visualization and



design validation, it lacks fully integrated event management features such as vendor coordination, task scheduling, and inventory management, limiting its functionality as a comprehensive event planning solution.

E3D Planner

E3D Planner is a web-based platform that provides instant 3D event mapping through an intuitive drag-and-drop interface, supporting collaboration and cross-device compatibility. The system emphasizes rapid workflows, allowing planners to design a 3D event within minutes and generate renders in seconds, thereby reducing revisions and accelerating client approvals. Its feature set includes 4K photorealistic renders, reusable templates, smart seating and guest management, and inventory tracking, all accessible via desktop or mobile devices. However, while it excels in speed, visualization, and interactive planning, it lacks comprehensive integration with broader event management functions, such as vendor coordination, budget tracking, and task assignment, which limits its capacity as a fully centralized event management solution.

3D Event Designer

3D Event Designer is a web-based platform that enables event planners to create accurate 2D and 3D floor plans across any device desktop, tablet, or mobile without requiring downloads. The system provides eye-level and aerial 3D walkthroughs, reusable templates, and a comprehensive catalog of inventory items, including furniture, décor, and tents, allowing planners to produce realistic client visuals and expedite approval processes. The platform also supports embedding plans, sharing interactive views, generating seating charts, and exporting

artifacts for vendors, facilitating seamless collaboration among stakeholders. However, while it excels in immersive visualization and layout planning, it lacks integrated event management features such as vendor coordination, task scheduling, and collaborative editing, which limits its effectiveness as a fully comprehensive event planning solution.

CHAPTER III

RESEARCH METHODOLOGY AND TECHNICAL BACKGROUND

This chapter describes the research methodology and technical framework used in the development of the Mobile and Web-based Event Layout Planner with 3D Visualization and Preview. It presents the research approach, design, and methods applied in the study, including data collection procedures, population and sample selection, research instruments, and research processes used to gather information from relevant stakeholders.

METHODS AND TECHNIQUES OF THE STUDY

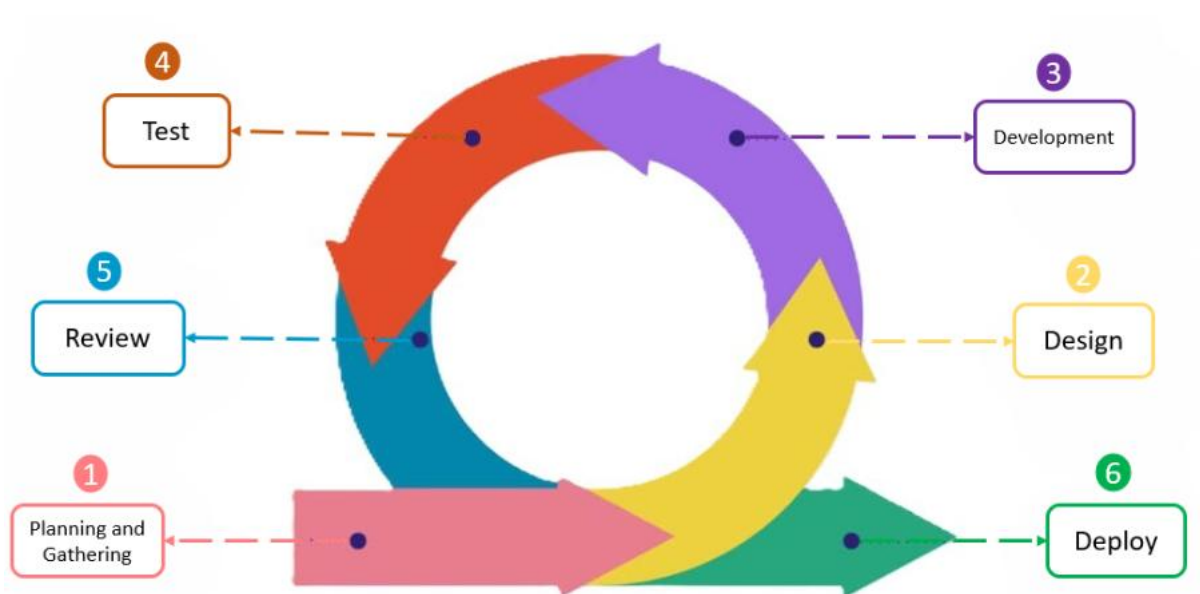


Figure 1: *Agile Methodology*

The researchers utilize Agile Methodology as the primary framework for this study. Agile is an iterative and incremental approach to software engineering that prioritizes flexibility, rapid delivery, and continuous improvement through stakeholder collaboration. Unlike traditional "Waterfall" models where each phase must be completed before moving to the next, Agile allows the researchers to develop the system in small functional pieces, ensuring that the 2D/3D visualization and AI features are tested and refined throughout the entire development process (Beck et al., 2001).

The study follows an Iterative-Developmental Research Design. This design integrates data collection and software development into a continuous lifecycle, allowing the researchers to adapt to new requirements or technical challenges as they arise. This methodology ensures that the Mobile and Web-based Event Layout Planner remains user-centered and technically sound from the initial prototype to the final version.

To implement this methodology, the researchers employ the following techniques:

- Iterative Requirement Gathering - Rather than defining all features at once, the researchers conduct ongoing interviews and observations with event planners and clients. This allows for the continuous discovery of necessary features for the venue management and scheduling modules.
- Incremental Development - The system is built in several versions or "increments." For example, the basic 2D layout tool is developed first, followed by the integration of 3D visualization, and finally the AI-assisted design and cost computation features. Each increment adds more value and complexity to the system.

- Continuous Feedback and Testing - At the end of every development cycle, quantitative surveys and usability tests are performed with end-users. This data-driven feedback loop enables the researchers to identify bugs, improve the user interface (UI), and ensure that the layout previews are accurate.
- Technical Evaluation - Throughout the development cycles, the system is subjected to experimental evaluation by IT experts. This ensures that the architectural integrity, security, and cross-platform synchronization (between web and mobile) meet the ISO/IEC 25010 quality standards.

By utilizing an Agile approach, the researchers ensure that the final product is not just a completed software package, but a solution that is rigorously tested, highly adaptable to the needs of the event industry, and technically optimized for mobile and web platforms.

POPULATION AND SAMPLE OF THE STUDY

The population of the study consists of stakeholders involved in event planning and management, including individuals who use or manage event planning services and professionals responsible for evaluating digital systems. This includes 15 administrators who manage event planning businesses, 15 customers who employs the firm's event services, 15 event organizers who plan and execute events, 15 venue administrators who manage event venues, and 5 IT experts responsible for evaluating the technical functionality of the system. Including multiple stakeholder groups ensures that the system addresses both practical and technical needs, supporting usability, relevance, and overall effectiveness. According to Uldall-Espersen and Frøkjær (2007), usability during software development must consider

both user interface and organizational perspectives, and the continued engagement of stakeholders, who carry the vision of the project, is critical to achieving system goals. In addition, research in mobile service development emphasizes that involving potential stakeholders in the design and development process is vital for achieving the best service experience. Methodologies such as User-Centered Design (UCD) ensure that stakeholders have direct input, guiding both functional and non-functional aspects of system design (Ammattikorkeakoulu, 2015). Together, these findings support the inclusion of diverse stakeholder groups in the study to provide meaningful and relevant insights for system evaluation.

Purposive sampling will be used to select participants for this study. This non-probability sampling technique is chosen to focus on participants who have direct experience with event planning and digital systems. Purposive sampling allows the study to gather meaningful feedback from those who are most knowledgeable and capable of providing insights relevant to the development and evaluation of the system. According to Etikan (2016), purposive sampling is particularly useful when randomization is not possible, when the population is large, or when the study does not aim to generalize results to the entire population, making it an appropriate choice for targeting information-rich respondents. The actual sample of the study is composed of selected respondents based on their expertise and level of involvement in event planning or system evaluation. Event admins and event organizers provided input on practical usability and workflow; customers provided feedback on user experience; venue administrators evaluated venue management and system integration; and IT experts assessed technical functionality, usability, and system performance. The number of respondents in each group will be determined based on expertise requirements and usability

testing guidelines, which indicate that small but targeted groups can detect the majority of interface and functionality issues while larger numbers improve the reliability of evaluations.

According to Alroobaea and Mayhew (2014), empirical studies on usability evaluation methods show that small groups of participants are often sufficient to uncover most usability problems, while larger groups can further enhance reliability.

Table 1.

Sample Distribution of Respondents

Stakeholder Group	Role in the Study	Population Description	Number of Respondents
Event Company Admins	To evaluate business processes, system usability, and decision-support features of the event layout planner.	Owners or managers of event planning firms who oversee event operations and business workflows.	15
Customer	To assess user experience, interface usability, and satisfaction with the system.	Individuals who avail of event planning services and interact with the system for viewing and selecting event layouts.	15
Event Organizer	To evaluate event planning, execution, and layout visualization features, while providing feedback on operational workflow, coordination, and system usability.	Professionals responsible for the end-to-end lifecycle of an event, from initial planning and schedule management to final implementation. In this study, they evaluate the system's practical utility in coordinating client requirements and executing layout designs.	15

Venue Administrator	To assess venue-related data management, system integration, and layout compatibility.	Individuals responsible for managing event venues, facilities, and venue specifications.	15
IT Expert	To evaluate technical functionality, system performance, and usability compliance.	Information technology professionals with expertise in system development, evaluation, and maintenance.	5
TOTAL			65

Note. Table 1 shows the Population Group and Size of Study of EventScape.

The distribution of respondents will be determined based on their expertise and relevance to the mobile and web-based event layout planner with 3D visualization and preview. Event planning firm admins, coordinators, and event organizers were included to ensure that business workflows, planning processes, and execution requirements are accurately reflected in the system design. Customers were selected to provide feedback on user experience, interface usability, and overall satisfaction with the system. Venue administrators contributed insights related to venue management, layout compatibility, and system integration, while IT experts evaluated the technical functionality, system performance, and usability compliance. The number of respondents per group are carefully selected to balance the need for specialized, information-rich feedback with practical considerations of conducting surveys, interviews, and usability testing, thereby ensuring a comprehensive evaluation of the system's functionality, usability, and effectiveness.

RESEARCH INSTRUMENT

The questionnaire is designed to gather quantitative data on the usability, efficiency, reliability, and user satisfaction of the developed mobile and web-based event layout planning system with 3D visualization and preview. It is divided into three sections: respondent demographics, system evaluation, and feature assessment.

The demographics section included questions about the participant's role, level of experience, and familiarity with event planning or system usage. This information is used to generate the frequency and percentage distribution of respondents in the succeeding chapter. Respondents were selected through **purposive sampling**, a **non-probability sampling technique** in which participants are deliberately chosen based on specific criteria relevant to the study's objectives. This method is most appropriate when the study requires respondents who possess particular knowledge, experience, or characteristics directly aligned with the research focus, as the identification of appropriate sampling units must be in line with the research questions and scope of the study (Memon et al., 2025). This approach ensured that only individuals with direct experience or involvement in event planning, venue management, or system use were included as evaluators, thereby producing informed and contextually relevant responses. The items for usability, efficiency, reliability, and satisfaction were guided by the **ISO/IEC 25010:2023 software quality model** to ensure alignment with internationally accepted software quality characteristics. A **four-point Likert scale** will be employed for responses, where 4 – Strongly Agree, 3 – Agree, 2 – Disagree, and 1 – Strongly Disagree, providing standardized, comparable data across respondent groups.

For the qualitative portion, **open-ended questions** are included to explore respondents' experiences with the system, challenges encountered during navigation or task completion, and recommendations for improvement. These questions allowed participants to elaborate on their insights, providing rich qualitative data that complemented the quantitative survey results. Additionally, system evaluation forms are employed during usability testing to record observations regarding system performance, responsiveness, navigation, and error occurrences. Respondents tested critical features such as event layout customization, 3D visualization, preview, and navigation, while researchers documented their experiences to identify areas for refinement. To ensure validity, reliability, and accuracy, all instruments were reviewed by expert evaluators and approved by the researchers' thesis adviser before data collection. By combining these quantitative and qualitative tools, the study can collect comprehensive and reliable data for evaluating the interaction capability, functional suitability, and flexibility of the proposed system.

Table 2.

Research Instrument Guidelines Based on ISO/IEC 25010:2023

Factors	Description (Relevance to EventScape)
Functional Suitability	Ensures the system provides all essential functions including real-time 2D and 3D visualizations, booking management, and financial cost computations to fully meet the needs of both web and mobile users.
Performance Efficiency	Assesses the system's responsiveness and resource utilization when rendering real-time visualizations and processing booking transactions across mobile and web platforms, especially under high-load conditions.
Compatibility	Measures how well the web-based dashboard and mobile application integrate and synchronize with each other, as well as with external databases and third-party services involved in booking and payment processing.

Interaction Capability	Focuses on the ease of navigation, accessibility, and consistency of the user interface across both the mobile application and web dashboard, ensuring users can interact with 2D/3D tools intuitively.
Reliability	Guarantees the system remains operational and recovers from failures without losing critical booking records or financial computation data, maintaining availability across both platforms.
Security	Protects sensitive user booking data and financial transactions through encryption and resistance to unauthorized access, ensuring integrity and confidentiality throughout the system.
Maintainability	Evaluates how easily the system's codebase and database structure can be analyzed, modified, and updated to support new features, platforms, or evolving visualization requirements.
Flexibility	Assesses the system's adaptability and scalability across different devices, operating environments, and deployment configurations, ensuring consistent performance as the user base grows.
Safety	Ensures the system operates within defined constraints to prevent critical failures such as data loss during booking transactions or computation errors that could negatively impact users or financial integrity.

Note. Table 2 shows the ISO/IEC 25010:2023 guidelines for quality characteristics

DATA GATHERING AND PROCESSING PROCEDURE

For the data-gathering procedure, the researchers employed qualitative methods through direct client engagement and expert consultation to gather the necessary information for the design and development of the EventScape system. Given that the study is currently ongoing, the data-gathering activities conducted thus far focused on establishing the system's foundational requirements and ensuring that the development direction aligns with both the client's operational needs and the academic standards set by the researchers' adviser.

Prior to the conduct of the study, the researchers coordinated with their group adviser to obtain guidance on the appropriate documentation and direction of the thesis. The adviser provided recommendations regarding the system's features, technical specifications, and documentation requirements, which the researchers incorporated into the ongoing development and refinement of the study. Consultations with the adviser were conducted on a continuing basis throughout the research process to ensure that revisions and improvements to both the system and the thesis paper remained aligned with academic expectations.

To gather the primary data necessary for defining the system's functional requirements, the researchers conducted meetings and interviews with the client. During these sessions, the client provided information regarding their operational workflows, event management practices, venue layout needs, and the specific features they required from the proposed system. The researchers documented the client's responses through written notes and audio recordings taken during the sessions, which were subsequently reviewed and organized to extract key data points relevant to the system's design and development. The information gathered from the client served as the primary basis for determining the system's core functionalities, user roles, scheduling requirements, and layout management specifications.

All data obtained from the client meetings and adviser consultations were treated with strict confidentiality. The recorded responses and notes were securely stored and accessed exclusively by the research team for the purpose of system development and thesis documentation. As the study remains ongoing, further data-gathering activities, including system evaluation and validation procedures, will be conducted in subsequent phases of the research in accordance with the approved research methodology.

PROJECT TECHNICAL DESCRIPTION

EventScape is a mobile and web-based Event Layout Planner with 3D Visualization and Preview developed to support event organizers, affiliated venues, and clients in designing, managing, and visualizing event layouts. The system provides a centralized digital platform that enables authorized users to create and customize 2D/3D layouts with 3D previews, manage event schedules and venue assignments, generate cost estimates, and export layouts in PDF format. Using AI-assisted structural analysis and machine learning-driven conflict detection, EventScape integrates intelligent design tools with a practical and accessible cross-platform interface. The mobile application serves Event Organizers, Venue Administrators, and Clients, while the web application provides Event Company Admins with full administrative oversight. The system is hosted online using Hostinger web hosting and Hostinger VPS services, allowing users to securely access the platform through internet-connected devices at any time.

1. Programming Languages and Technologies

The development of EventScape utilizes a modern, decoupled software architecture to ensure cross-platform compatibility, responsiveness, maintainability, and efficient performance across both mobile and web environments.

Architectural Layer	Specific Technology	Specification / Version	Purpose in the Project
Frontend Framework	Expo / React Native	Expo 54, Expo Router 6, React 19	Powers the unified, mobile-first routing structure and native cross-platform UI.



3D Rendering Core	Three.js Engine	@react-three/fiber, @react-three/drei, expo-gl	Drives the interactive 3D canvas, asset positioning, object collision boundaries, and dynamic extrusion.
Client Web Build	React Native Web	Standard Expo Web target compilation	Compiles the shared application components into responsive, static web browser code.
Application Backend	FastAPI	Python Runtime with Uvicorn & Pydantic	Runs the dedicated server-side microservice hosting heavy data workflows and parsing pipelines.
Traditional Vision API	OpenCV	opencv-python-headless, NumPy	Handles deterministic image decoding, grayscale thresholding, and vector line tracing.
Cloud Infrastructure	Supabase	@supabase/supabase-js	Governs cloud data persistence, secure user authentication session states, and relational database tables.
Media Repository	Cloudinary	Signed parameter uploads via Edge Functions	Hosts profile avatars and image assets securely without loading down the application database.
External Security API	Google reCAPTCHA	WebView integration & Edge Function verification	Provides security challenges and verification tokens for authentication abuse protection.
External Location API	Google Maps API	Google Places & Geocoder browser integrations	Facilitates visual map display and geographical venue address lookups.
On-Device Location	Expo Location	Native foreground permission module	Captures device permission status and reverse-geocodes audit metadata.

Frontend Technology

The frontend of the system is developed using Expo and React Native (Expo 54, Expo Router 6, React 19), which powers the unified, mobile-first routing structure and native cross-platform user interface. React Native Web is additionally used under the standard Expo Web target compilation to compile the shared application components into responsive, static browser-compatible code. This unified approach allows the system to maintain a single codebase deployable across Android, iOS, and web environments.

3D Rendering

Three-dimensional visualization is handled by the Three.js engine through `@react-three/fiber`, `@react-three/drei`, and `expo-gl`. This rendering core drives the interactive 3D canvas, manages asset positioning and object collision boundaries, and performs dynamic extrusion of 2D layout geometries into full 3D representations matching real-world venue dimensions.

Backend Technology

The backend of the system runs on a FastAPI microservice within a Python runtime environment powered by Uvicorn and Pydantic. FastAPI handles server-side processing, heavy data workflows, AI model inference pipelines, and API communication between the frontend clients and server-side computational resources. The language and framework were selected for their performance efficiency, native support for asynchronous processing, and compatibility with machine learning libraries.

Computer Vision Processing

Traditional computer vision tasks are performed by OpenCV (opencv-python-headless with NumPy), which handles deterministic image decoding, grayscale thresholding, and vector line tracing. OpenCV serves as the geometry engine that processes neural network predictions into precise, topology-safe room graphs used for layout extrusion and structural analysis.

Cloud Infrastructure and Storage

Cloud data persistence, user authentication session management, and relational database operations are governed by Supabase through the @supabase/supabase-js client library. Media assets such as profile avatars and uploaded images are hosted via Cloudinary using signed parameter uploads through Edge Functions, ensuring secure asset delivery without overloading the application database.

Security and Location Services

Authentication security is reinforced through Google reCAPTCHA, integrated via WebView and Edge Function verification, which provides bot prevention and verification token handling for user authentication flows. Geographical venue address lookups and map display are facilitated by the Google Maps API through Google Places and Geocoder browser integrations. On-device location services are managed by the Expo Location module for foreground permission capture and reverse-geocoding of audit metadata.

2. Artificial Intelligence and Machine Learning Models

Rather than relying on a single generalized AI model, EventScape distributes its machine learning architecture into isolated, task-specific components to maintain geometric accuracy and processing efficiency. Each AI model is assigned a bounded, specialized role to prevent interference between the geometric coordinate pipeline and natural language processing layers.

AI Model Family	Specific Model	Processing Location	Model Role and Purpose
Server-Side Vision Engine	Ultralytics YOLO11m (Fallback: Detectron2 / Mask R-CNN)	Server-Side GPU (FastAPI Backend)	A Deep Convolutional Neural Network (CNN) that analyzes uploaded floorplans to extract bounding boxes for doors, windows, and walls.
Local Vision / Geometry Engine	OpenCV + NumPy deterministic extraction	Server-Side FastAPI Backend	Generates the core wall graph from floorplan pixels. Produces topology-safe finalWalls, openings, windows, wall runs, host axes, and debug evidence.
Logistical Tree Ensembles	XGBoost & RandomForest	Server-Side CPU (FastAPI Backend)	Classical tabular models that process structured numeric features (guest count, room area, asset lists) to predict scheduling setup duration and flag layout positioning conflicts.
Metadata Processing LLM	Mistral 7B (or gpt-oss-20b)	Server Cloud Container	A lightweight, open-weight Large Language Model (LLM) used strictly to clean up messy text scanned from blueprints and format them into standard JSON templates.

Optional Mask Refiner	SAM / SAM2	Server-Side GPU (Selective Use)	Utilized as an annotation assistant or secondary refiner to resolve ambiguous geometry layouts or complex architectural symbols.
Optional Depth Estimation	Depth Anything V2 Small (or ZoeDepth)	Server-Side GPU (Photo Mode)	Provides monocular depth estimation to calculate scale and distance metrics when processing raw room photographs instead of flat blueprints.
Optional 3D Reconstruction	Instant-NGP (or Plenoxels)	Server-Side GPU (R&D Capture)	Used exclusively in dedicated venue photo/video capture modes for real-time multi-view radiance-field 3D reconstruction.

A. 3D Layout and Structural Intelligence (Neural Networks)

Instead of relying on end-to-end black-box networks to generate 3D layouts blindly, EventScape employs a hybrid approach combining Deep Convolutional Neural Networks (CNNs) and computer vision heuristics across three stages:

- **Semantic Localization:** Valid floorplan image uploads are transmitted to the server-side FastAPI service, which invokes an Ultralytics YOLO11m model (yielding 51.5 COCO mAP with 20.1M parameters) or an Apache-2.0 compliant Detectron2 / Mask R-CNN engine to predict spatial bounding boxes and masks for critical structural entities including walls, doors, windows, and structural fixtures.
- **Deterministic Graph Compilation:** Rather than promoting raw neural network inferences directly into permanent floorplan geometry, the pipeline processes these predictions as semantic hypotheses. A deterministic geometry engine maps these inferences against linear pixel arrays using OpenCV to build precise axis hypotheses, calculate straight-line wall segment extensions, and construct a topology-safe, editable room graph.

- **3D Spatial Extrusion:** The finalized, validated 2D layout coordinates are ingested by the frontend React Three Fiber / Three.js scene, which dynamically extrudes the vector lines into 3D solid geometries, matching real-world dimensions while avoiding object-clipping or safety clearance overlapping.

B. Event Scheduling, Capacity, and Conflict Scoring (Classical Tabular ML)

For operational logistics and business rules, the system avoids Large Language Models entirely and utilizes classical machine learning tree ensembles to handle structured tabular variables:

- **Predictive Asset Scheduling and Duration Estimation:** Using features such as guest count, physical venue area, layout clutter ratios, and equipment requirements, a RandomForest or XGBoost regressor calculates realistic scheduling metrics, predicting venue setup and teardown velocities and optimizing slot allocation.
- **Host-Wall Trust Verification:** Traditional machine learning classifiers inspect generated layout candidates, scoring properties such as overlap ratios, structural span support, and OCR text agreement to ensure the system rejects false detections before altering the user's workspace geometry.
- **Scale and Performance Optimization:** For expansive corporate events with dense data logs, LightGBM provides a speed-optimized, lightweight alternative to run fast training iterations without increasing server compute latency.

C. Metadata Parsing and Taxonomy Normalization (LLM Layer)

Large Language Models are restricted exclusively to natural language workflows, deliberately insulating the geometric coordinate core from text hallucinations. The system uses an open-weight, Apache-2.0 licensed Mistral 7B or gpt-oss-20b model for OCR post-processing, ingesting chaotic text tokens extracted from blueprint scans, stripping noise, reformatting erratic labeling configurations, and mapping room designations into clean, uniform JSON schemas.

3. Deployment, Portability, and Compliance Strategy

EventScope follows a multi-target deployment strategy designed to ensure cross-platform portability, online accessibility, and academic evaluability.

Web Application Target

The frontend web application is compiled into optimized, responsive, standalone static assets including HTML, CSS, and production-ready JavaScript bundles. These static files are deployed and hosted on Hostinger Shared Web Hosting, ensuring fast global content delivery, high uptime, and minimal client-side browser loading overhead.

Mobile Application Target

The mobile application code is compiled into native iOS and Android packages leveraging the Expo application framework. The compiled mobile binaries are deployed to the Google Play Store for Android users and the Apple App Store for iOS users, targeting their respective native operating system environments.

Academic Evaluation

For prototype validation, developer testing, and formal academic evaluation, the native system is simultaneously bundled as a standalone Android APK binary hosted via cloud storage. Reviewers can download and install this binary immediately onto physical testing devices by scanning a system-generated QR code download link.

Edge Inference Engine

On-device model processes such as MobileNetV3 run locally on the client machine via ONNX Runtime Mobile and LiteRT. This ensures that local image validation logic remains cross-platform and executes natively over iOS CoreML or Android XNNPACK drivers, allowing on-device quality gating to function without a mandatory internet connection.

Hosted Server-Side Core

Because the heavy vision pipelines (YOLO11m) and classical machine learning tree structures (XGBoost) require a dedicated Python runtime environment and continuous computational processing, this layer is containerized and deployed inside a specialized FastAPI web server instance hosted on a Hostinger VPS (Virtual Private Server). The microservice is configured with custom Cross-Origin Resource Sharing (CORS) security controls to securely handle incoming data streams from the authorized web and mobile frontends.

ETHICAL CONSIDERATION

The development and evaluation of the Mobile and Web-based Event Layout Planner with 3D Visualization and Preview involve the collection and processing of personal and organizational data across multiple user roles. The integration of artificial intelligence, cloud infrastructure, and location services within the system further reinforces the researchers' responsibility to uphold ethical standards. This study adheres to established legal frameworks and ethical principles to protect participant rights, ensure data privacy, and promote the responsible use of technology throughout the research process.

Compliance with Data Privacy Laws (Republic Act No. 10173)

The study strictly complies with the Data Privacy Act of 2012 (Republic Act No. 10173), the primary Philippine law governing the protection of personal information in information and communications systems. This law requires that all personal data be collected for specified and legitimate purposes, processed lawfully, and retained only for as long as necessary (Republic Act No. 10173, 2012). In the context of EventScape, this applies to all data gathered during account registration, event scheduling, venue assignments, and layout management. All user data is collected with the knowledge and consent of the data subject and is safeguarded against unauthorized access or disclosure in accordance with the Act.

Alignment with International Data Protection Standards

Beyond local compliance, the study aligns with internationally recognized data protection principles, particularly the General Data Protection Regulation (GDPR) of the European Union, which emphasizes data minimization, purpose limitation, and storage limitation. Since EventScape utilizes cloud infrastructure through Supabase and Cloudinary,

responsible data governance across these platforms is necessary. Studies on cloud-based compliance have emphasized that organizations must implement structured, layered data governance approaches including continuous monitoring throughout the data lifecycle to minimize the risk of unauthorized data exposure (Olawale et al., 2024).

Privacy-by-Design and Data Security

EventScape integrates a Privacy-by-Design approach into its system architecture, ensuring that privacy and security mechanisms are built into the system from the earliest stages of development. The GDPR formally mandates Privacy-by-Design, requiring that data protection be implemented at the system design stage and by default across all processing activities (Cambronerio et al., 2024). For EventScape, this includes encrypted data storage and transmission, secure authentication reinforced by Google reCAPTCHA, and role-based access controls that restrict system access according to each user's designated role. Role-Based Access Control (RBAC) has been recognized as an effective and scalable security mechanism for managing user permissions in multi-user digital platforms, granting fine-grained access rights while reducing exposure risks (Singh, Rani, & Kumar, 2024).

Informed Consent and Voluntary Participation

All participants involved in the evaluation of the system provide informed consent prior to any engagement with the platform or submission of personal information. Participants are fully informed about the study's purpose, the data being collected, how it will be used, and their right to withdraw at any time without consequence. This is consistent with ethical guidelines emphasizing that the consent process must clearly communicate study details and participant rights to support fully informed, autonomous decision-making (McInnis et al., 2024).



Participation is entirely voluntary, and alternative respondents will be sought if any party declines involvement.

Governance of Artificial Intelligence

The integration of AI and machine learning models within EventScape, including YOLO11m for structural layout detection, XGBoost and RandomForest for scheduling conflict analysis, and Mistral 7B for OCR metadata normalization, carries specific ethical obligations regarding transparency and accountability. Each AI component is limited to a specific, well-defined task and functions strictly as a decision-support tool rather than an autonomous decision-maker, ensuring that human oversight is preserved throughout all AI-assisted workflows. This aligns with the principle that incorporating human oversight in AI decision-making is critical to accountability, especially in systems where automated outputs directly affect user data and organizational workflows (Kashefi et al., 2024). Transparency in how AI systems operate is equally essential to ensuring responsible deployment and user trust (Cheong, 2024).

Intellectual Property and Content Ownership

All event layouts, venue templates, and user-generated content created within EventScape remain the exclusive intellectual property of the respective users or organizations that produced them. The researchers claim no ownership over any content generated through the platform. Any content included in academic reporting is done only with explicit consent from the owning party. This is consistent with the principle that users who create and upload content to digital platforms are the rightful copyright holders of their work, and any use of such content must be governed by clear consent and transparent terms (Fashion Law Journal, 2024).



Transparency, Confidentiality, and Ethical Oversight

All data collected during the evaluation of EventScape are used exclusively for academic research and system improvement. No data is shared with third parties or used for commercial purposes, and results are reported in aggregate form to preserve participant's anonymity. Protecting participant information is both a legal and ethical obligation, requiring researchers to restrict access to identifiable data and commit to non-disclosure in published outputs (Wolf et al., 2015). All research instruments, evaluation procedures, and ethical safeguards in this study were reviewed and approved by the researchers' thesis adviser prior to implementation to ensure compliance with academic research ethics and Philippine data privacy regulations.

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